



Question Set

Question 1

Consider a collateralized forward, FRA, or vanilla swap desk position. Define the state variables (discount factors, forwards, accrual factors, and payoff dates) and derive the present-value and par-rate relation. State the measure, filtration, and tradability assumptions.

$$V_0 = E^Q \left[e^{-\int_0^T r_t dt} X_T | \mathcal{F}_0 \right]$$

Question 2

For a collateralized forward, FRA, or vanilla swap desk position, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in a collateralized forward, FRA, or vanilla swap desk position. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using a semiannual five-year swap curve with spot zero rates between 3.3 percent and 4.2 percent, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

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Question 5

For a collateralized forward, FRA, or vanilla swap desk position, compute and interpret PV01, forward-rate delta, and convexity correction. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes deposit, futures, and swap par-rate quotes. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for a collateralized forward, FRA, or vanilla swap desk position. Explain how linear payoff maturity values and discount-factor limits prevents economically impossible prices.

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Question 8

Design a Monte Carlo or lattice estimator for a collateralized forward, FRA, or vanilla swap desk position. Specify scenario shocks to the curve and forward level, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements a collateralized forward, FRA, or vanilla swap desk position and gets a stable but wrong answer. Explain how double-counting discounting or applying a par-rate formula to non-par cashflows creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for a collateralized forward, FRA, or vanilla swap desk position. Include inputs, outputs, diagnostics, and one benchmark case.

$$V_0 = \mathbb{E}^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

Question 11

Starting from the payoff definition of a collateralized forward, FRA, or vanilla swap desk position, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of a semiannual five-year swap curve with spot zero rates between 3.3 percent and 4.2 percent. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for a collateralized forward, FRA, or vanilla swap desk position. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in a collateralized forward, FRA, or vanilla swap desk position. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for a collateralized forward, FRA, or vanilla swap desk position is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for deposit, futures, and swap par-rate quotes are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

$$V_0 = \mathbb{E}^Q \left[e^{-\int_0^T r_t dt} X_T \mid \mathcal{F}_0 \right]$$

Question 17

For a collateralized forward, FRA, or vanilla swap desk position, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for scenario shocks to the curve and forward level. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that double-counting discounting or applying a par-rate formula to non-par cashflows could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

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Question 20

Explain a collateralized forward, FRA, or vanilla swap desk position and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.